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QUESTION 7

7.1.1	$\sin(A+B)$ $=\cos[90°-(A+B)]$ $=\cos(90°-A-B)$ $=\cos[(90°-A)-B]$ $=\cos(90°-A).\cos B+\sin(90°-A).\sin B$ $=\sin A.\cos B+\cos A.\sin B$	✓ A co-ratio ✓ A re-arrangement ✓ A expansion	(3)
7.1.2	$\sin(2x+50°)-\sin 15°\cos 48°=\sin 48°\cos 15°$ $\sin(2x+50°)=\sin 48°\cos 15°+\sin 15°\cos 48°$ $\sin(2x+50°)=\sin 63°$ $2x+50°=63°+k.360°$ OR $2x+50°=180°-63°+k.360°$ $2x=13°+k.360°$ $2x=67°+k.360°$ $x=6,5°+k.180°$, $k\in Z$ $x=33,5°+k.180°$, $k\in Z$	✓A using compound angle identity ✓ A both solutions ✓CA $x=6,5°+k.180°$ ✓CA $x=33,5°+k.180°$ $k\in Z$: penalty of 1 if not written at least once	(4)
7.2.1	$\frac{4\sin x\cos x}{2\sin^2 x-1}$ $=\frac{2(2\sin x\cos x)}{-(1-2\sin^2 x)}$ $=\frac{2\sin 2x}{-\cos 2x}$ $=-2\tan 2x$	✓ A $2\sin 2x$ ✓ A $-\cos x$ ✓ CA answer	(3)
7.2.2	$2\sin^2 x-1=0$ $\sin x=\pm\frac{1}{\sqrt{2}}$ $x=-45°$ or $x=45°$	✓ A $\pm\frac{1}{\sqrt{2}}$ ✓ CA $-45°$; CA ✓ $45°$	(3)
7.2.3	$\frac{4\sin 15°\cos 15°}{2\sin^2 15°-1}$ $=-2\tan 30°$ $=-2\left(\frac{1}{\sqrt{3}}\right)$ $=\frac{-2}{\sqrt{3}}$ or $\frac{-2\sqrt{3}}{3}$	✓ A $-2\tan 30°$ ✓ A answer	(2)
			[15]